BIOCHEMISTRY CUMULATIVE EXAM

February 12, 1994

Write a complete enzymatic pathway for the synthesis of glutamate in a liver cell suspension, using aspartate as the sole source of both carbon and nitrogen. You may assume that other metabolites are present in small catalytic amounts. Which carbon atom(s) of glutamate will first become labeled if aspartate labeled with C-14 at carbon number 3 is used?

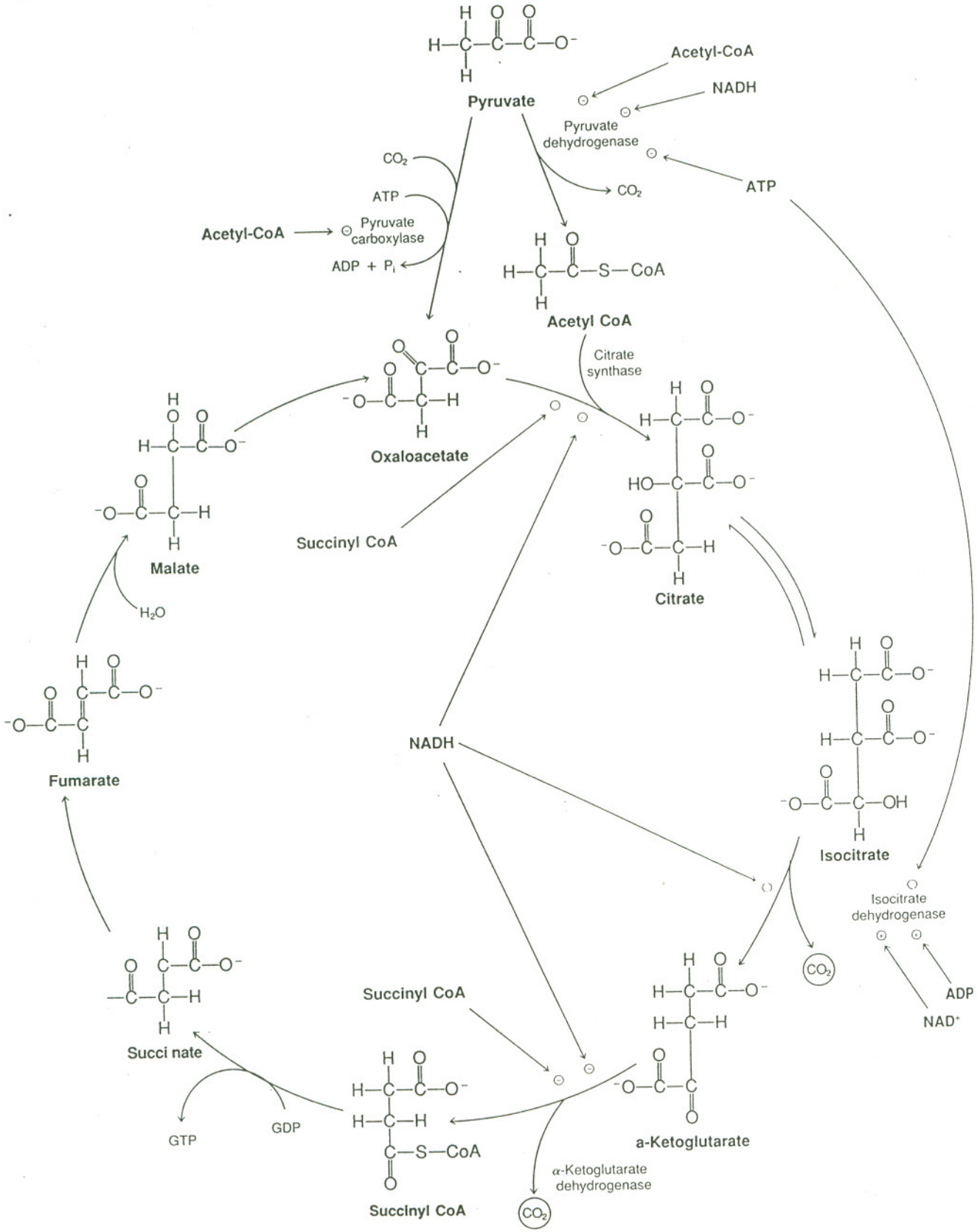

Figure 15–13
Major regulatory sites of the TCA cycle, with activators (⊕) and inhibitors (⊖) of specific reactions shown in color.

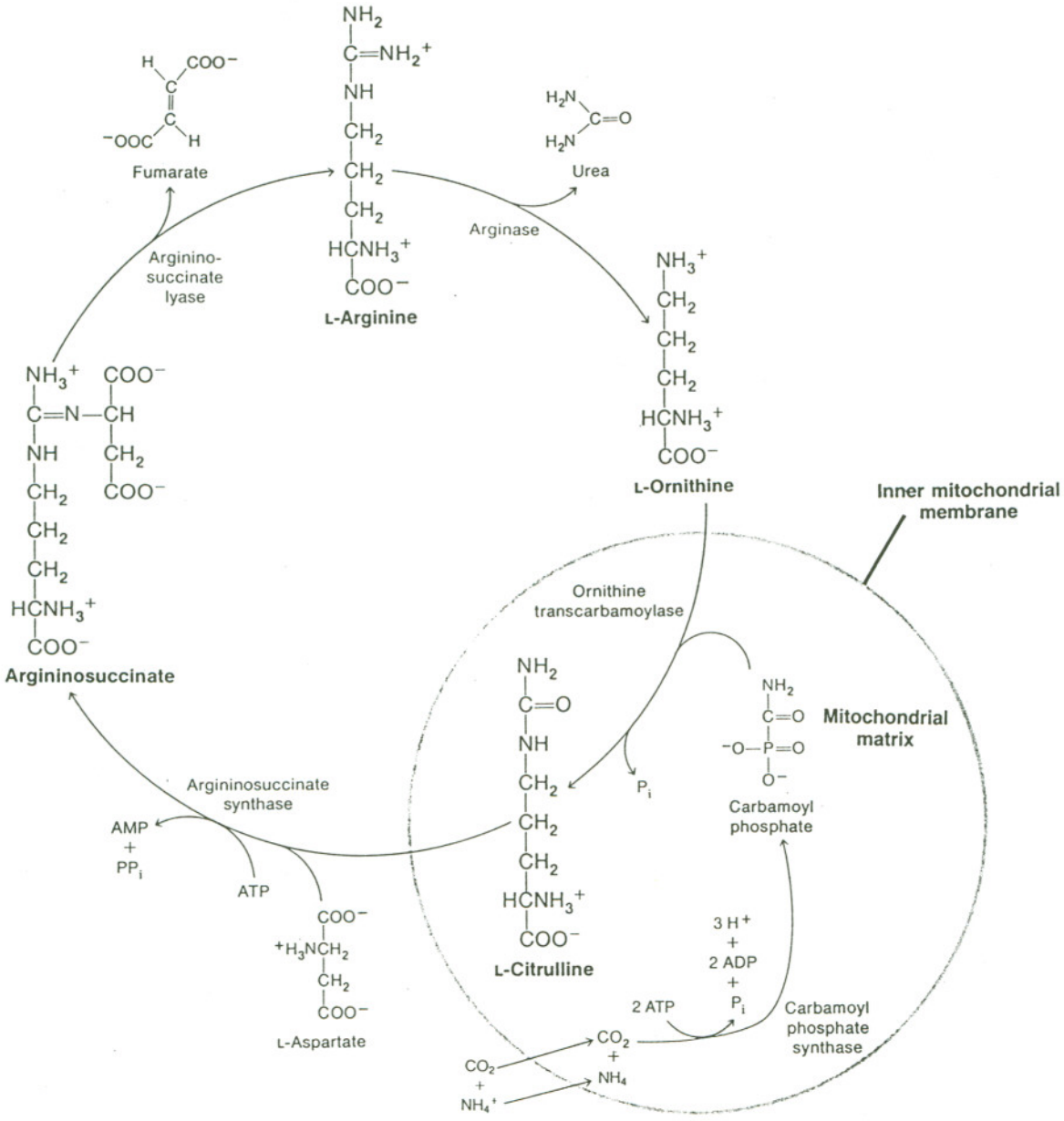
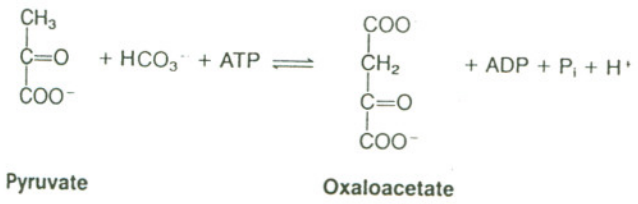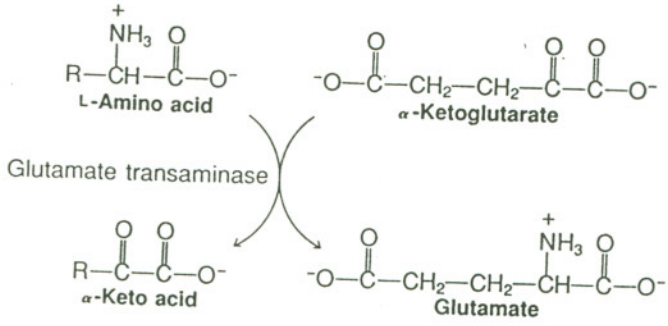
Figure 19–6
The urea cycle is a mechanism for removing unwanted nitrogen. Sources of nitrogens involved in urea formation are shown in color. Five enzymes are used in the urea cycle. Three of these function in the cytosol and two, as shown, function in the mitochondrial matrix. Specific carriers in the inner mitochondrial membrane transport ornithine, citrulline, ammonium ion, and HCO_3^- (CO_2) into and out of the mitochondrial matrix.